



INDUSTRIAL ATTACHMENT AT AKIJ BATHWARE LIMITED



Presented By

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INTRODUCTION

Akij Bathware Ltd., a subsidiary of Akij Group, launched ROSA, a premium brand of faucets and bathware, offering a combination of luxury and innovation. The brand was introduced to the market during a grand event on 11th October 2022, showcasing its high-quality products to industry leaders and dealers



What They Offer?



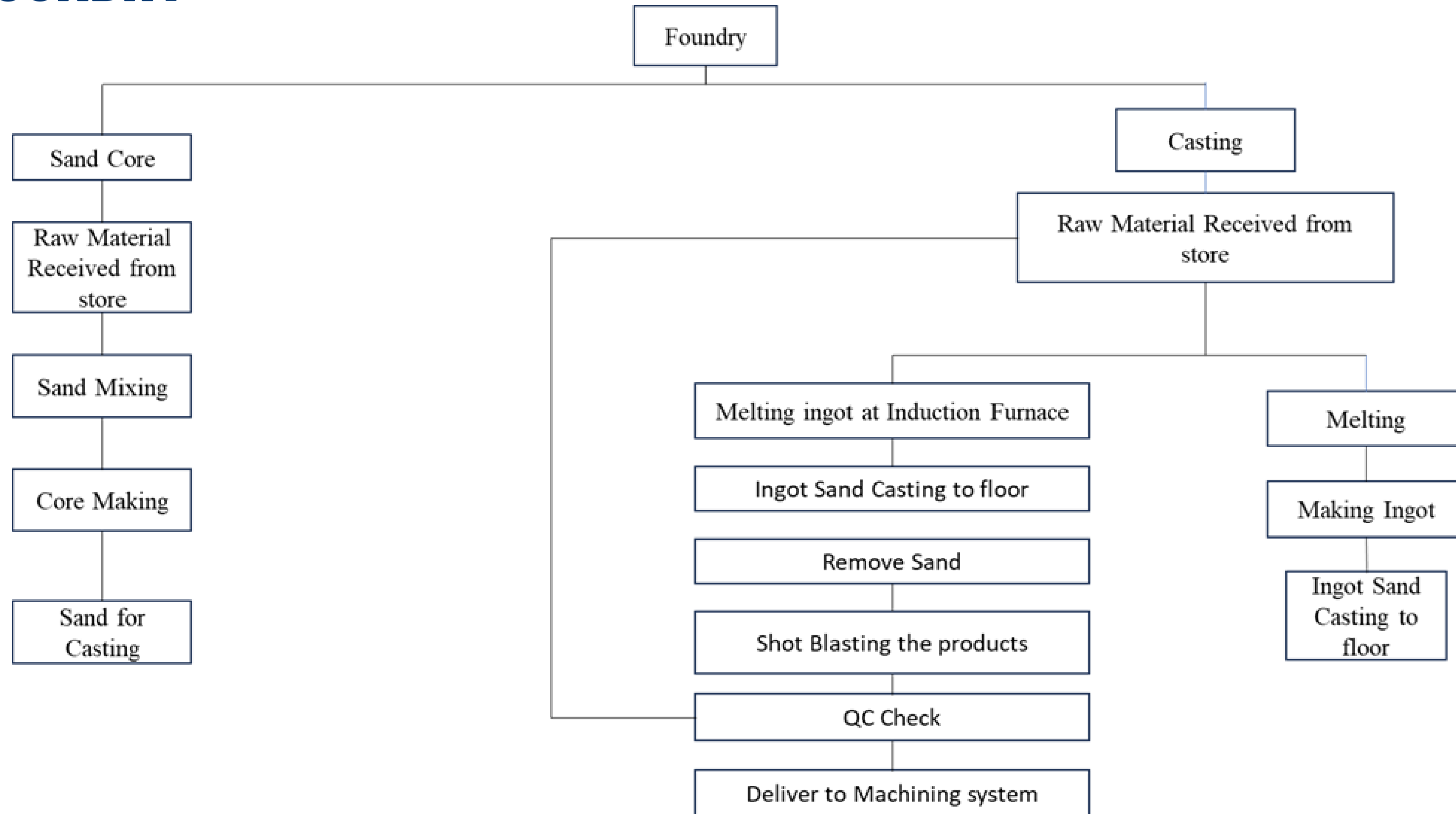
PRODUCTION BREAKDOWN



5 key Section



FOUNDRY





FOUNDRY

Sand Core Making Process



Silica Sand

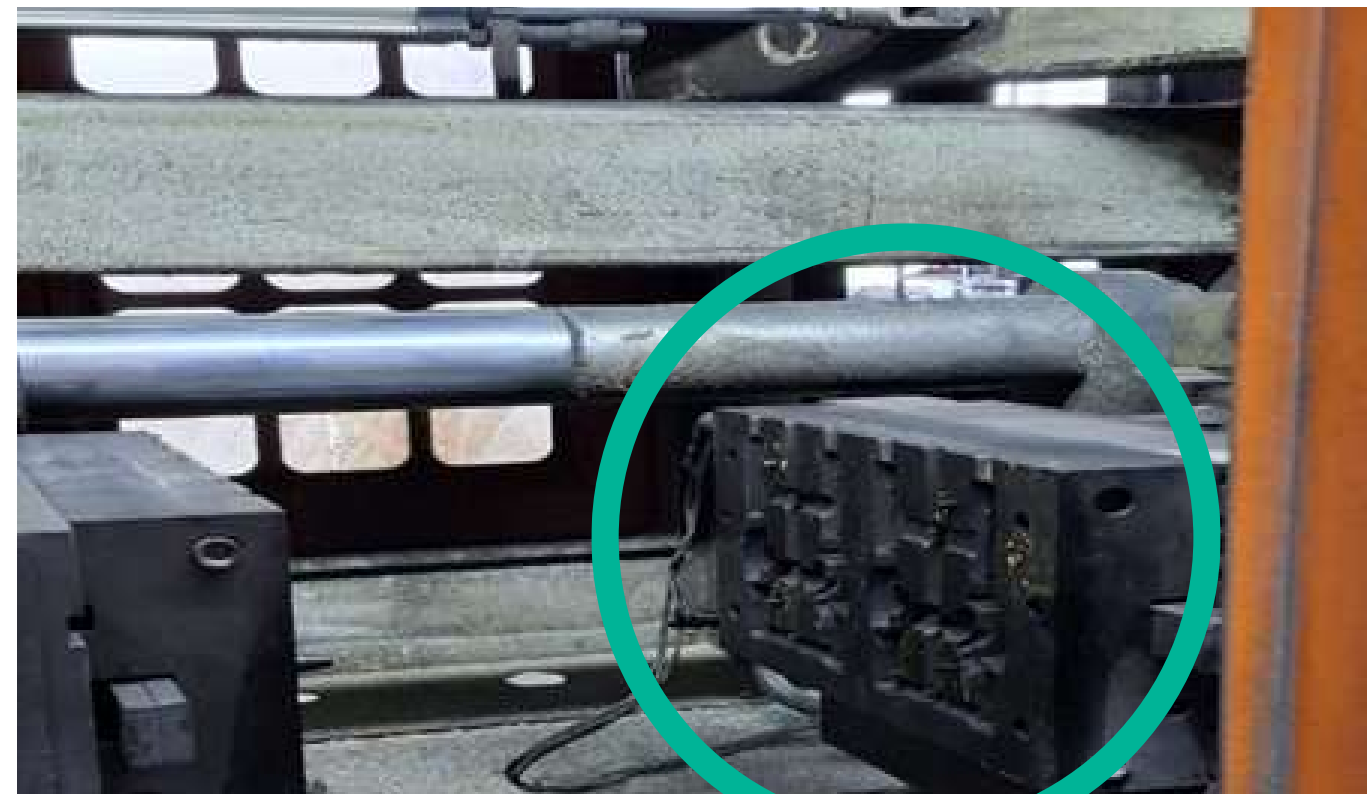


Resin



Binders

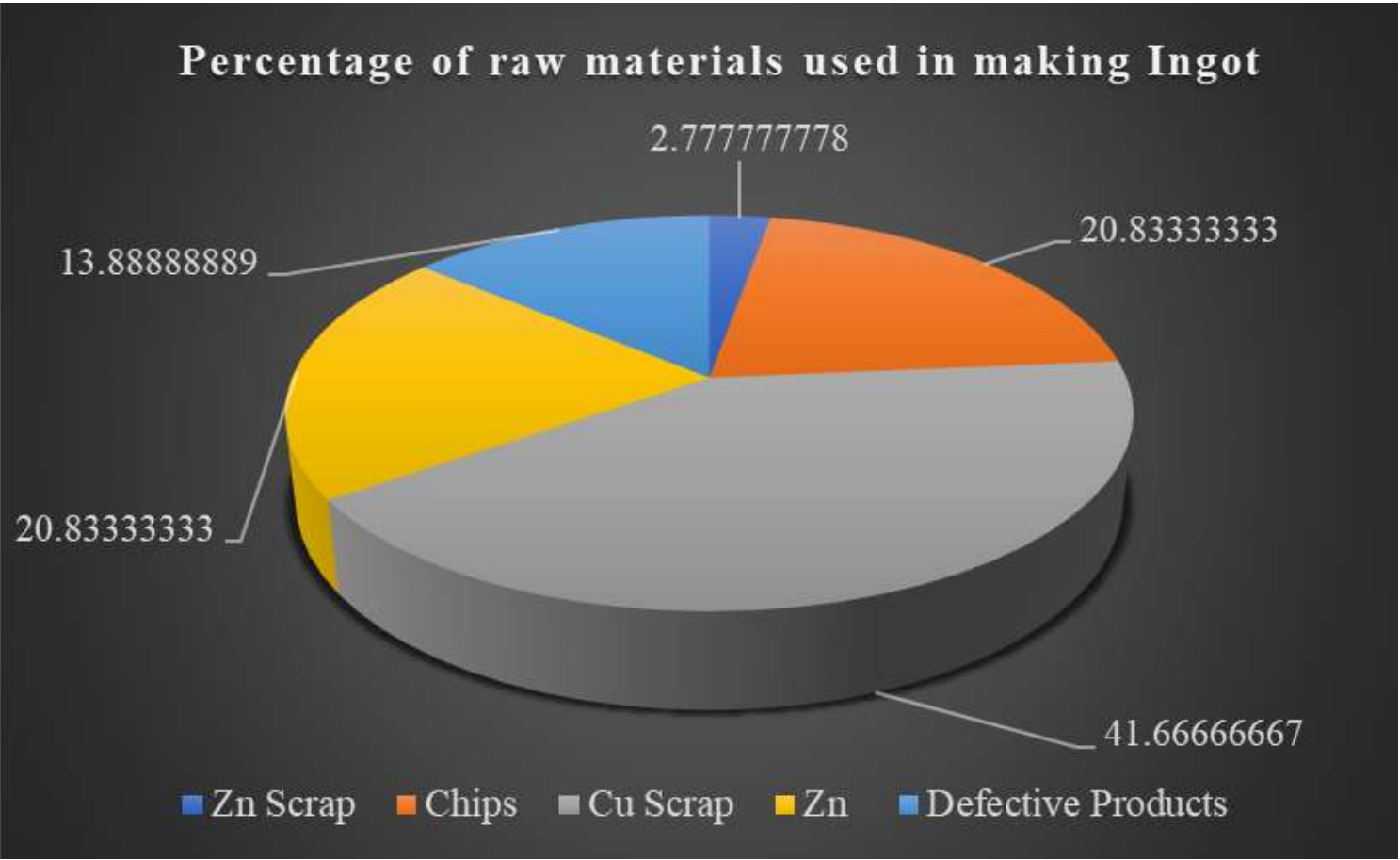
Sand Core Making summary Table	
Description	Value
Silica Sand	25 kg
Resin	700 gm
Shuntner	0.2 kg (200 gm)
Production of a particular core item in a day	250-300 Unit
Pressure used in Core Machine	6.5-7 kg/cm ²
Number of Automatic Core Machine	8
Core Making Time per Core	1.5 - 2 minutes per core



Ingot Making Process



Material	Type	Quantity(k	Percentage
Zn Scrap	Scrap	10	2.77777777
Chips	Machining	75	20.8333333
Cu Scrap	Scrap	150	41.6666666
Zn	metal	75	20.8333333
Defective	Recycled	50	13.8888888
Total		360	100



FOUNDRY

Casting

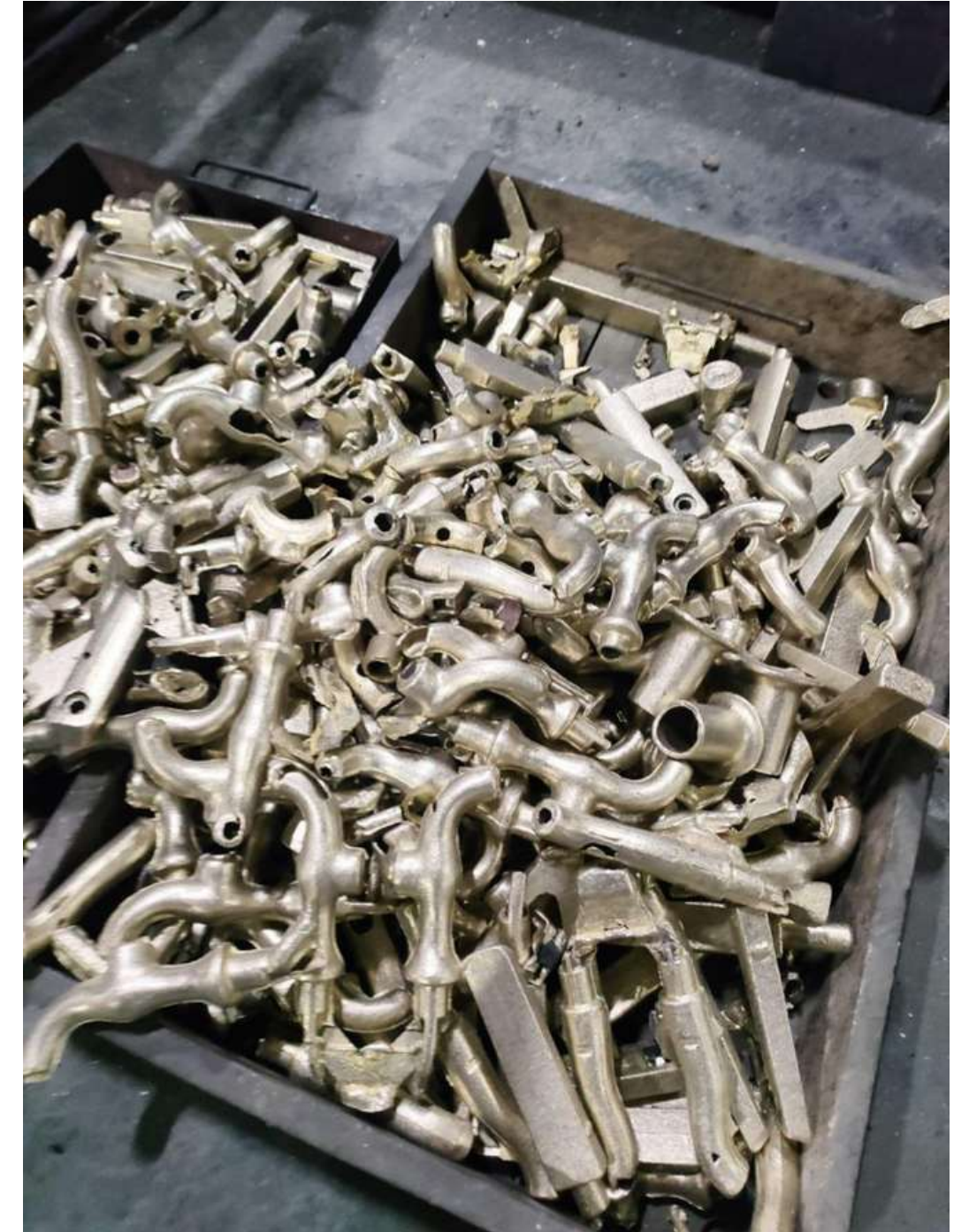


Table: Summary Table for Machines

Process	Machine type	Total Machines Available	Machines in Operation
Casting	Gravity Die Casting	4	4
	Low Pressure Die Casting	1	0
Melting	Induction Furnace	4	1

FOUNDRY

Casting



FOUNDRY

Casting Defects

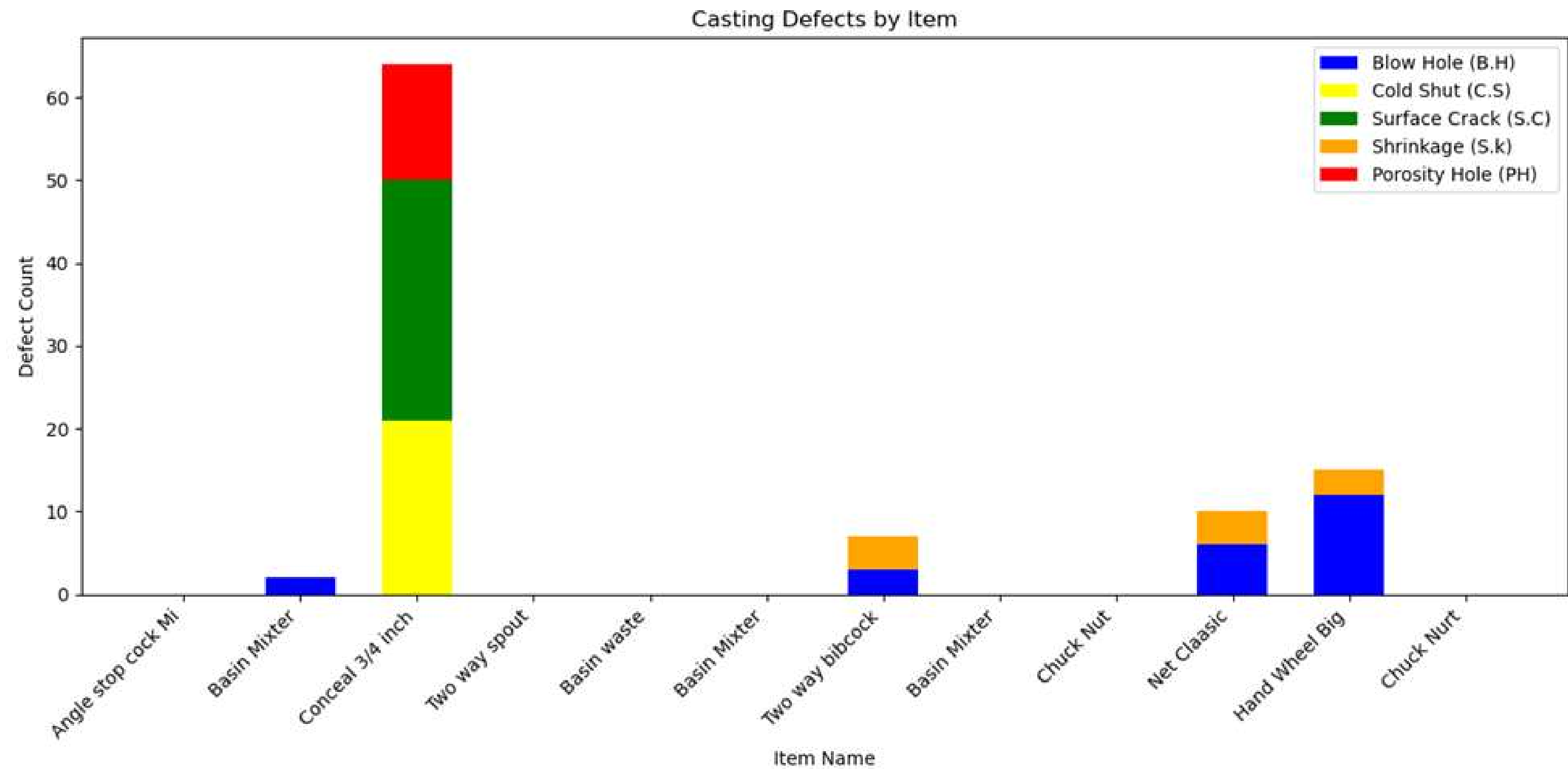


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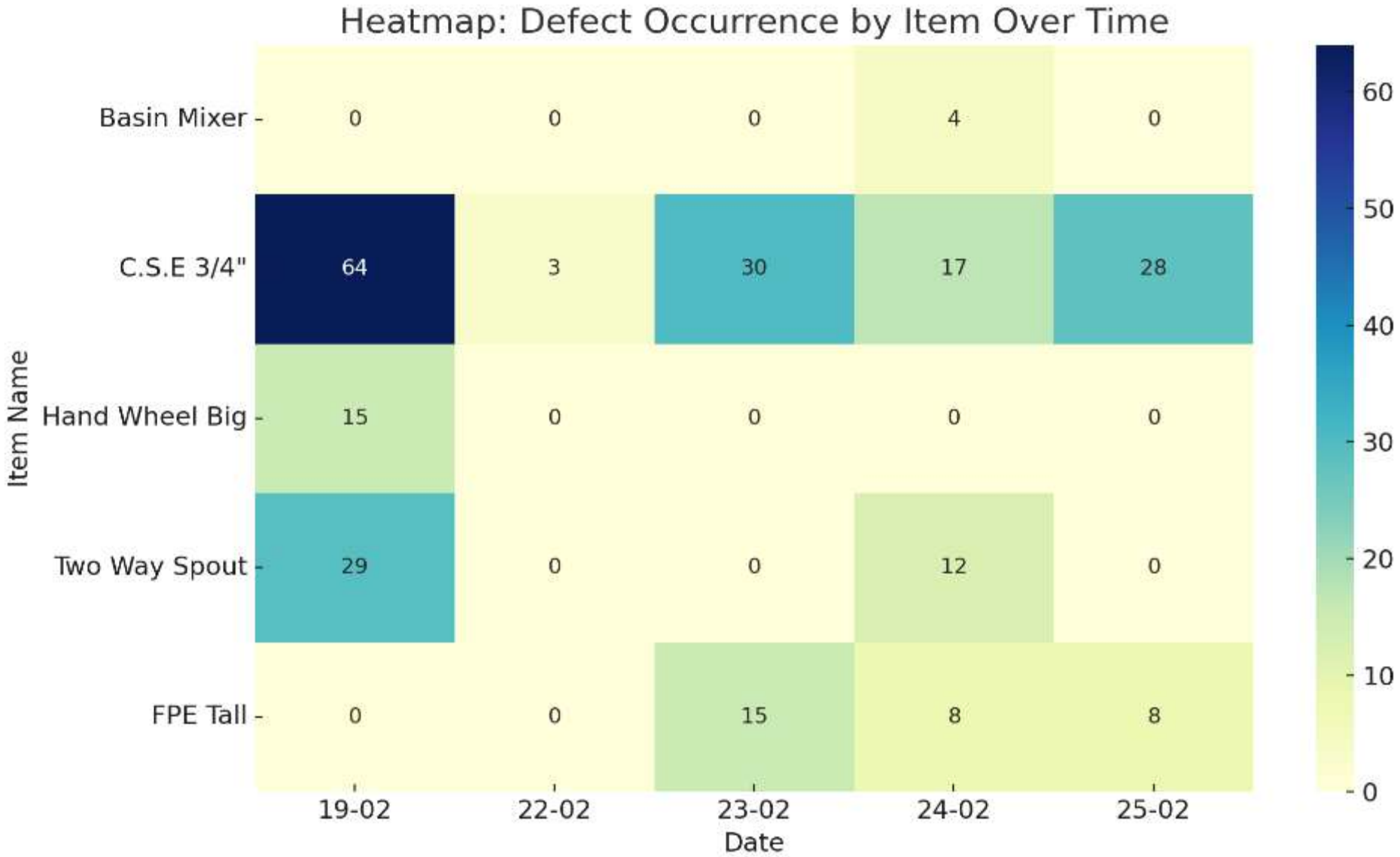
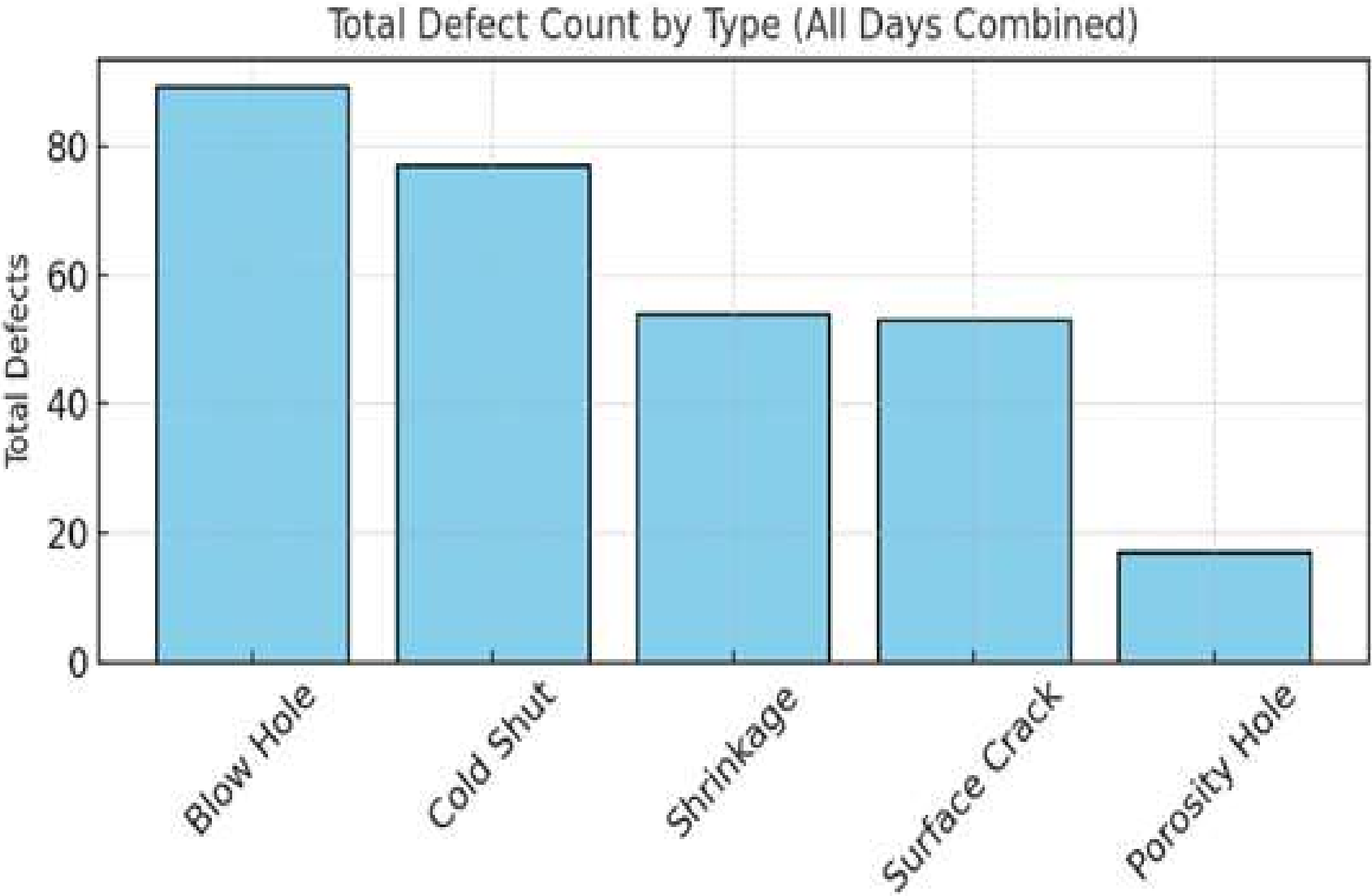
Casting Defects Analysis

Casting Summary of a Particular Day (19-02-2025)									
Code	Item Name	Making	Good	Rejected	Cause				
					Blow Hole (B.H)	Cold Shut (C.S)	Surface Crack (S.C)	Shrinkage (S.k)	Porosity Hole (PH)
3011	Angle stop cock Mini	566	566	0					
330	Basin Mixer	152	150	2	2				
803	Conceal 3/4 inch	334	270	64		21	29		14
4	Two way spout	1272	1243	29					
16	Basin waste	610	610	0					
470	Basin Mixer	410	410	0					
103	two way bibcock	541	534	7	3			4	
770	Basin Mixer	140	140	0					
701	Chuck Nut	1100	1100	0					
19	Net Claasic	730	720	10	6			4	
201	Hand Wheel Big	1415	1400	15	12			3	
16	Chuck Nurt	820	820	0					

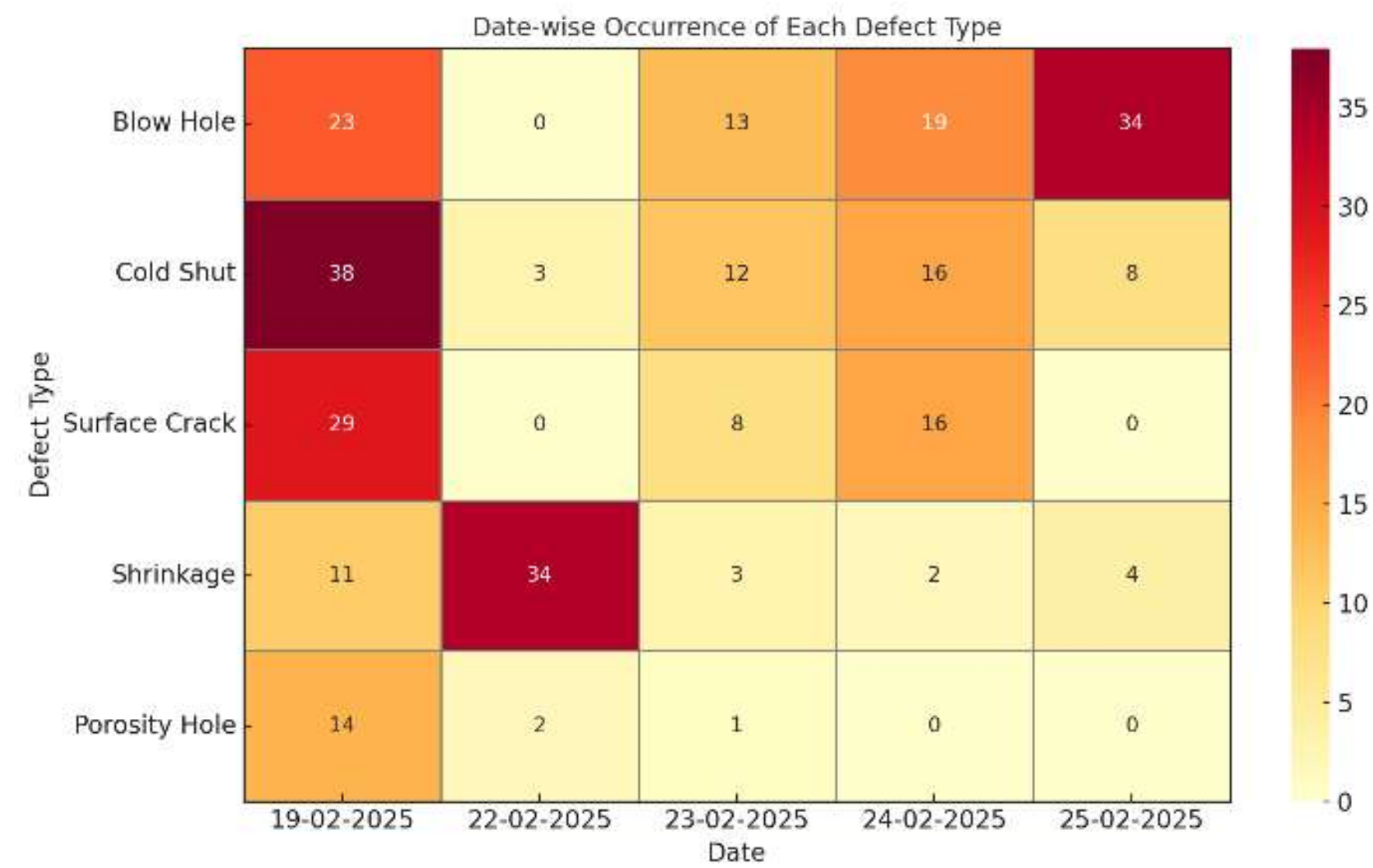
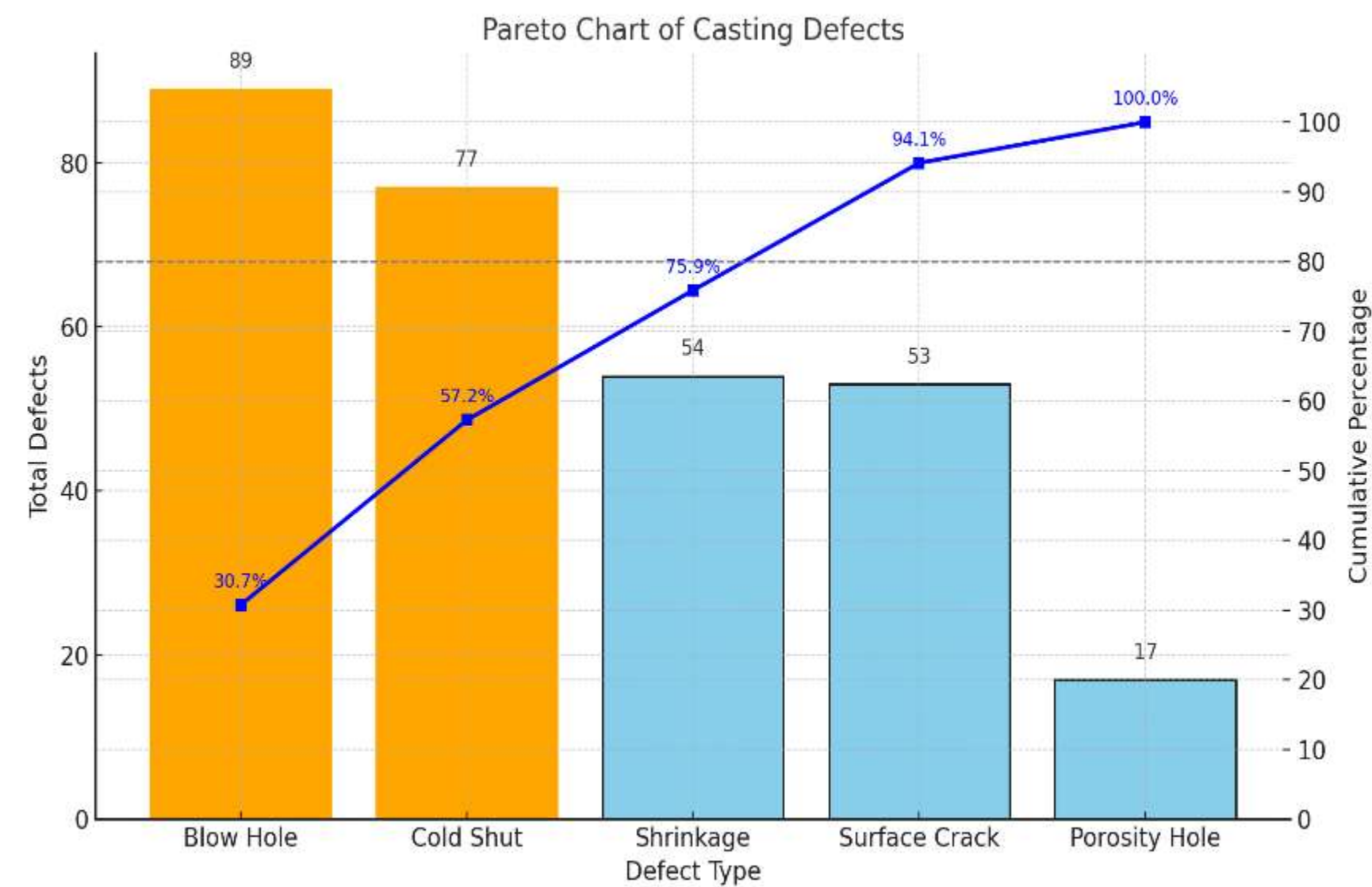
Casting Defects Analysis



Casting Defects Analysis



Casting Defects Analysis



Machining Section



What is Machining??

✓ Machining is the process of removing unwanted material for desire shape and size.

Summary of Machines



HMC Machine



VMC Machine



8 Spindle CNC



Manual Lathe



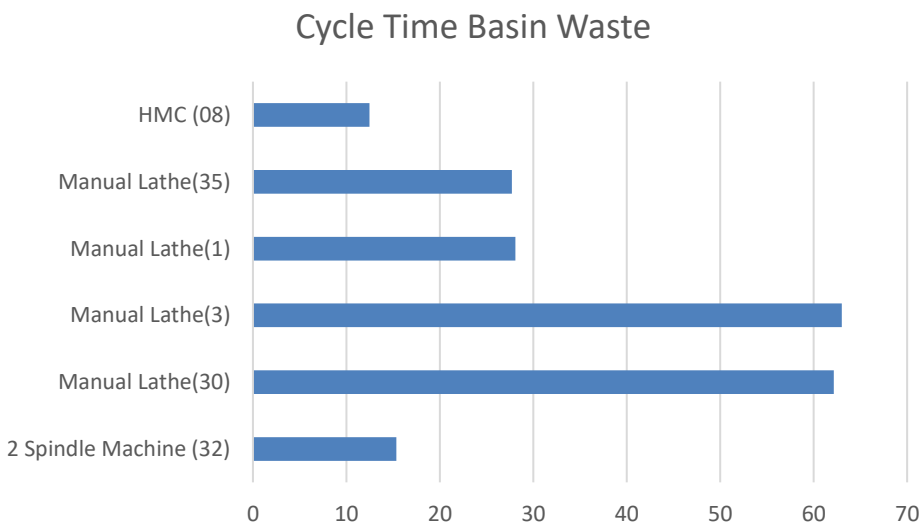
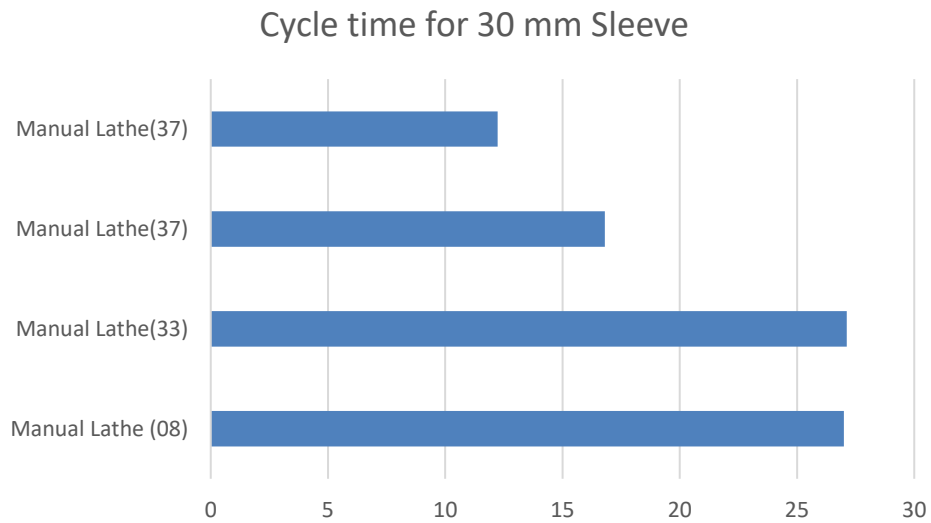
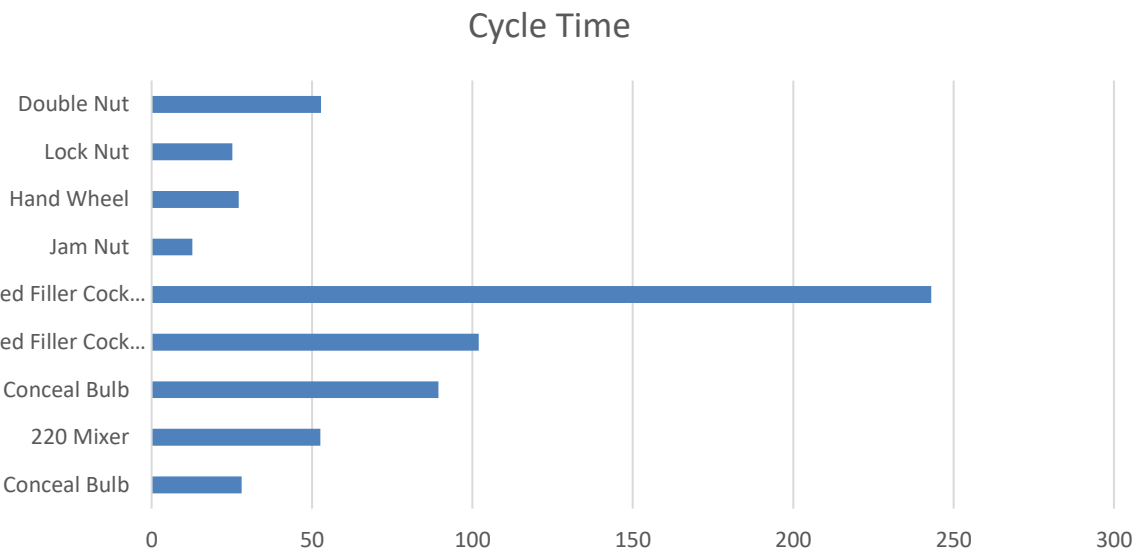
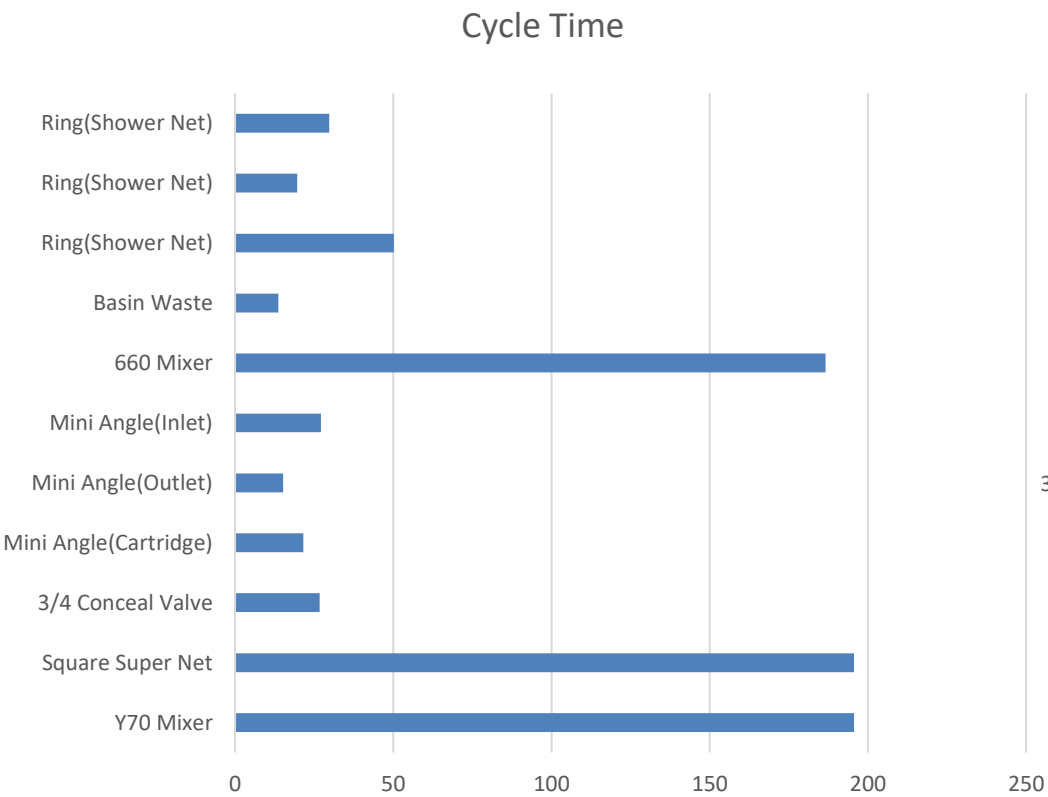
6 Spindle

Machining Section



Time Study in the Machining Section

SL. No	Product Name	Machine Name	Machining Process	Cycle Time	
	1Y70 Mixer	VMC Machine 04	Thread cutting, Boring, Facing	195.6Seconds	
	2Square Super Net	3 Spindle CNC	Side Cutting, Drilling	195.6Seconds	
	33/4 Conceal Valve	Six Spindle	Boring, Thread Cutting	26.76Seconds	
	4Mini Angle(Cartridge)	2 Spindle Machine	Thread cutting, Boring, Facing	21.6Seconds	
	5Mini Angle(Outlet)	2 Spindle Machine	Thread cutting, Boring, Facing	15.23Seconds	
	6Mini Angle(Inlet)	2 Spindle Machine	Thread cutting, Boring, Facing	27.12Seconds	
	7660 Mixer	VMC Machine 05	Thread cutting, Boring, Facing, drillind	186.6Seconds	
	8Basin Waste	2 Spindle	Facing	13.63Seconds	
	9Ring(Shower Net)	Capstan Lathe Machine(01)	Thread cutting, Boring, Facing	50.2Seconds	
	10Ring(Shower Net)	Manual Lathe Machine(35)	Boring	19.6Seconds	
	11Ring(Shower Net)	Manual Lathe Machine(34)	Turning	29.8Seconds	



Product: Basin Waste				
S.L No.	Machine Name	Process	Time Required for 5 Samples(Minutes)	Cycle Time
	12 Spindle Machine (32)	Facing	1.28	15.36Seconds
	2Manual Lathe(30)	Boring(Outlet)	5.18	62.16Seconds
	3Manual Lathe(3)	Thread Cutting(Outlet)	5.25	63Seconds
	4Manual Lathe(1)	Boring, Inside Facing	2.34	28.08Seconds
	5Manual Lathe(35)	Turning 3 portion	2.31	27.72Seconds
	6HMC (08)	Capsule Cut	1.04	12.48Seconds
		Total		3.48Minutes

Key Fundings & Observations in the Machining Section After Time Study

SL No.	Observations	Key Points	Remedy
1	Machine Sorting System	Machines are arranged based on product-specific needs to optimize workflow and reduce movement time.	- Organize machines efficiently so that all required machines for one product are grouped together. - Reduce unnecessary movement and setup time.
2	Automated Machining for Increased Productivity	Manual machining slows down production and increases errors.	- Assign more machining work to HMC, VMC, CNC lathe, 6 spindle, and 8 spindle VMC for efficiency. - Reduce manual operations where possible.
3	Quality Control (QC) Taking Too Much Time	QC delays the production process due to lengthy manual inspections.	
4	Manpower Management During Shift Changes	Poor shift coordination affects productivity and causes delays.	- Implement structured shift handover protocols. - Conduct team meetings before shifts to align work objectives. - Optimize workforce scheduling.
5	Safety & PPE Compliance	Workers are not consistently using required PPE, increasing the risk of workplace accidents.	- Enforce strict PPE rules with regular safety checks. - Provide safety gear like gloves, glasses, and ear protection. - Conduct periodic safety training.
6	Continuous Improvement & Kaizen Implementation	Lack of systematic continuous improvement slows down process optimization.	- Conduct regular Kaizen meetings to encourage worker participation. - Perform continuous assessments to identify inefficiencies. - Reward employees for successful improvements.

Key Observations

- Machines are currently arranged by product-specific needs to streamline flow and cut down on unnecessary travel time.
- Manual machining operations are slowing production rates and increasing defect occurrences.
- Quality Control processes involve extensive manual inspections, causing significant production delays.
- Shift changes lack a formal handover process, leading to misaligned objectives and downtime between teams.
- Workers frequently neglect required personal protective equipment, elevating the risk of accidents.
- There is no structured system for continuous improvement or Kaizen, which hinders ongoing process optimization.

Grinding & Polishing



Existing Setup Summary

✓ Manual Grinding Machines: 19 unit

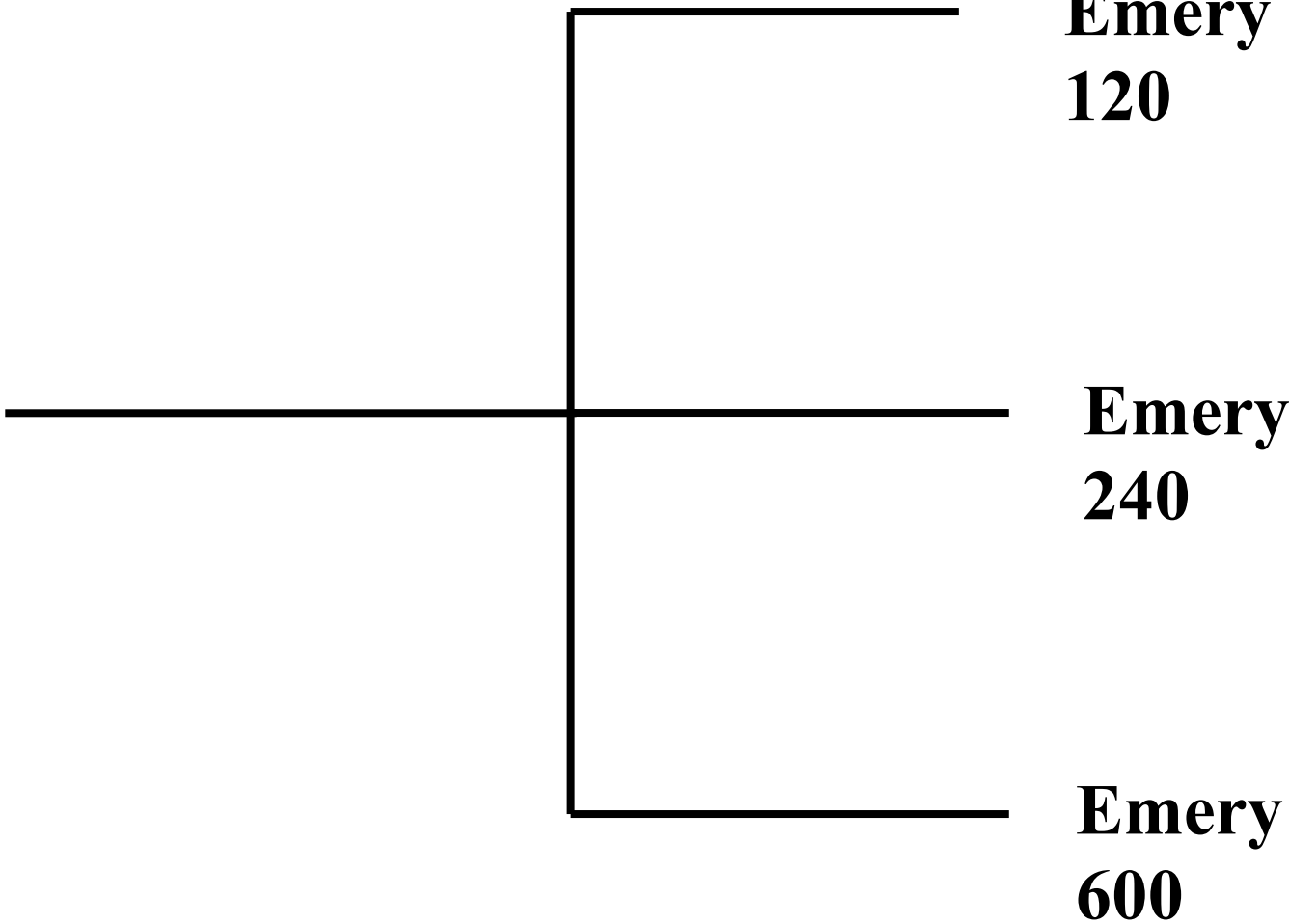
✓ Robotic Grinding Machine: 1 unit (grinds 4 products simultaneously)



Grinding & Polishing



3 Types of Emery papers are used here



Grinding & Polishing



After
Grinding



Before
Grinding

Basin mixture

Grinding & Polishing



Manual Buffing Polish Machines: 22

✓ 10 single-motor two-wheel machines



✓ 12 double-motor two-wheel machines



Grinding & Polishing



Current Machine Layout



Grinding & Polishing



A MAJOR ISSUE WAS OBSERVED



**Underutilization
of Machine !**

**High Idle Time on
Grinding Wheel**



Table. Cycle time and Takt Time analysis

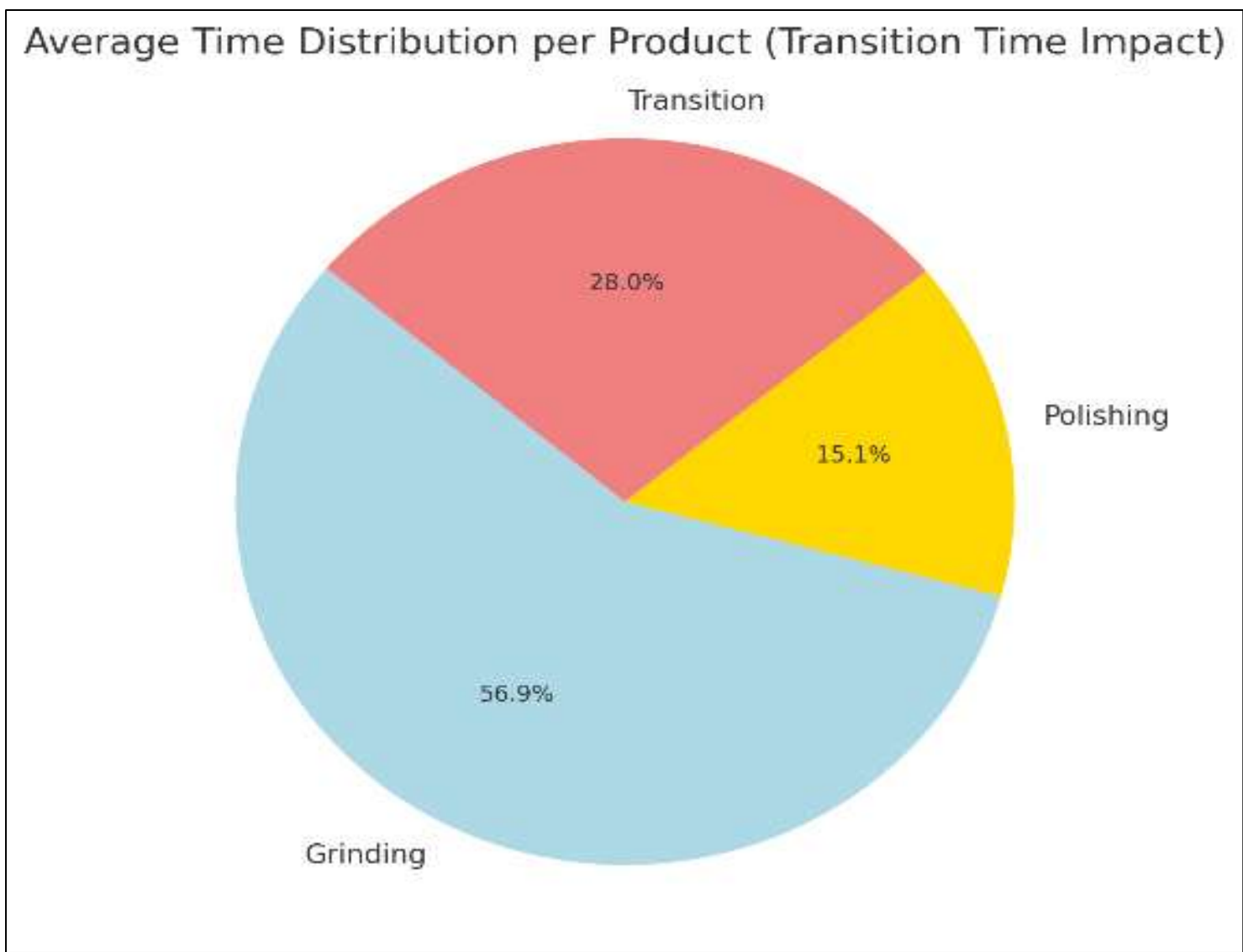
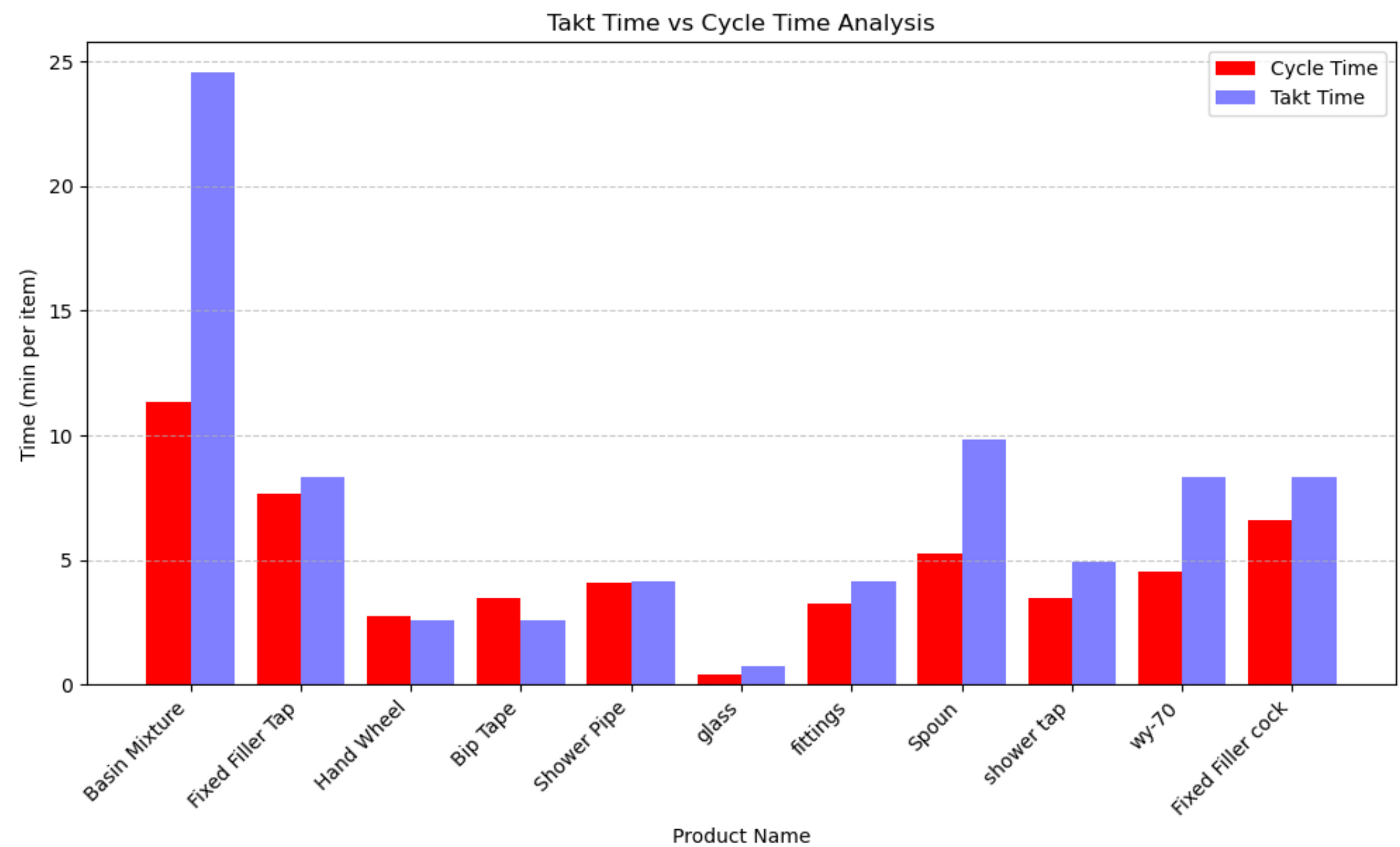
Product Name	Grinding Time (min)			Polishing Time (min)	Transition Time (min)
	Grinding Cycle Time using MRE-120	Grinding Cycle Time using MRE-240	Grinding Cycle Time using MRE-600		
Basin Mixture	5.3	3.49	1.33	1.24	1.58
Fixed Filler Tap	3.14	2.17	0.98	1.36	2.08
Hand Wheel	1.27	0.84	0.288	0.38	1.54
Bip Tape	1.08	0.92	0.61	0.87	2.12
Shower Pipe	1.46	1.19	0.92	0.52	1.46
glass	0.13	0.12	0.1	0.07	1.48
fittings	1.12	0.87	0.75	0.49	1.52
Spoun	1.49	0.95	0.77	2.07	2.31
shower tap	1.12	0.79	0.67	0.9	2.25
wy-70	1.47	1.15	0.7	1.19	2.16
Fixed Filler cock	2.24	1.38	0.97	2.01	2.02

Grinding & Polishing

Table. Cycle time and Takt Time analysis

Product Quantity (Demand)	Shift			Total Available Timen for Operation (hr)	Effective Operating Time (hr)	Takt Time (min per item)	Idle Time (hr)	Machine Nname	Cycle Time (per item in minute)
	Start Time	End Time	Overtime (hr)						
22	6:00	2:00	3	11	9	24.55	2.45	Grinding Machine-1	11.36
65	6:00	2:00	3	11	9	8.31	2.28	Grinding Machine-3	7.65
210	6:00	2:00	3	11	9	2.57	3.02	Grinding Machine-7	2.778
210	6:00	2:00	3	11	9	2.57	2.56	Grinding Machine-2	3.48
130	6:00	2:00	3	11	9	4.15	2.37	Grinding Machine-6	4.09
1500	6:00	2:00	3	11	9	0.36	3.04	Grinding Machine-13	0.42
130	6:00	2:00	3	11	9	4.15	2.32	Grinding Machine-8	3.23
55	6:00	2:00	3	11	9	9.82	2.39	Grinding Machine-4	5.28
110	6:00	2:00	3	11	9	4.91	3.12	Grinding Machine-12	3.48
65	6:00	2:00	3	11	9	8.31	2.58	Grinding Machine-2	4.51
65	0:00	2:00	3	11	9	8.31	2.52	Grinding Machine-6	6.6

Grinding & Polishing



Grinding & Polishing



We proposed 2 Solution

Change Current layout

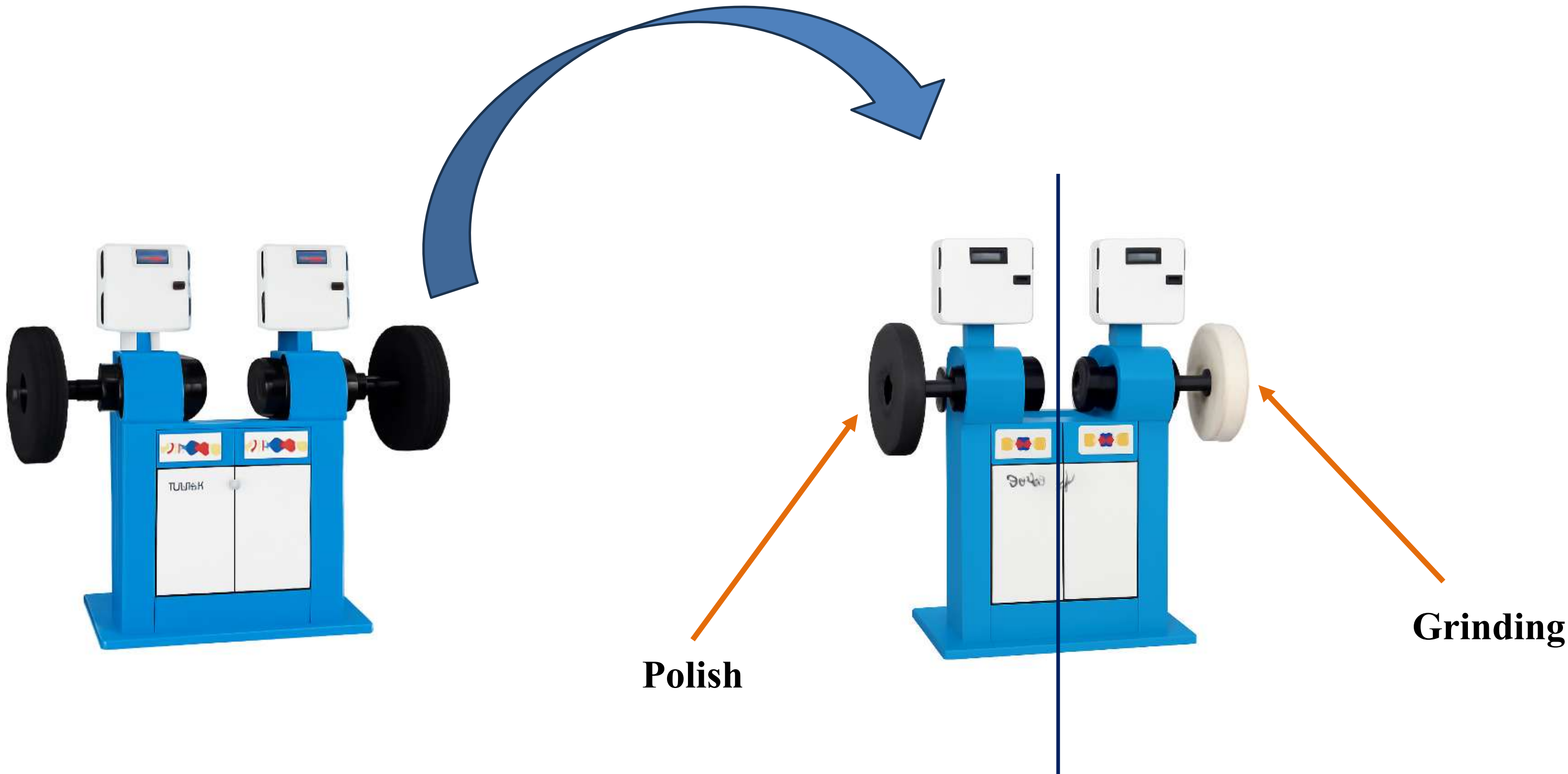
- Manual Grinding
- Manual Polish (Single Motor)
- Manual Polish (Double Motor)
- Auto Grinding
- Auto Polishing

Row 1	MG1	MP-S1	MG2	MP-S2	MG3	MP-S3	MG4	MP-S4	MG5	MP-S5	Auto Grinding	MP-D1	MG7	MP-D2	MG8	MP-D3	MG9
Row 2	MG10	MP-S6	MG11	MP-S7	MG12	MP-S8	MG13	MP-S9	MG14	MP-S10	Auto Polishing	MP-D5	MG16	MP-D6	MG17	MP-D7	MG18

Grinding & Polishing



Split Dual-Side Polishing Machines



Grinding & Polishing



Plan-Do-Check-Act (PDCA)

Date:		Team Member:		Department: Grinding and Polishing	
1: Plan	Problem	Identify Root Cause of Problem		Counter-Measures	
	1)High Machine Idle Time 2) One-Side Wheel Grinding Issue 3Improper Workflow & Role Allocation 4)Cycle Time Exceeds Takt Time 5)Bottleneck Formation 5)Layout & Workstation Organization.	Why?	Why does machine idle time occur?	Ans:	Option One
			Due to unbalanced task allocation and workflow inefficiencies.		Task Separation Between Grinding & Polishing
		Why?	Why is there unbalanced task allocation?	Ans:	
			Because one operator is responsible for both grinding and polishing, causing delays in switching between tasks.		
	Why?	Why does one person do both grinding and polishing?	Ans:	Option Two	
		Due to a lack of dedicated operators for each task, leading to inefficiencies		Machine Utilization Optimization	
	Why?	Why does cycle time exceed takt time?	Ans:		
		Because of bottlenecks created by improper workflow and frequent task switching			
	Goal	Why?	Why does improper layout cause delays?	Ans:	Option Three
1) Reduce machine idle time by 20% within 3 months 2) Balance workload distribution between grinding and polishing 3)Ensure grinding and polishing are performed by separate workers to reduce bottlenecks 4) Reduce cycle time to match or stay below		Due to the inefficient arrangement of machines and workstations, increasing travel time and reducing productivity.		Workstation Layout Redesign and Takt Time-Based Production Planning	
	Root Cause				
	The main causes of inefficiencies in the grinding and polishing section are poor task allocation, lack of dedicated operators, an unoptimized layout, and excessive machine idle time due to workflow imbalances.		Pros / Benefits: Rearrange workstations to reduce operator movement and improve flow and Adjust cycle time to match takt time by setting standard work sequences.		Cons / Challenges: May require temporary disruption in production and Needs continuous monitoring to ensure efficiency.

Grinding & Polishing



2: Do	Containment	Implementation Tasks for Long-Term Counter-Measure				Criteria to Determine Effectiveness
	1) Assign separate workers for grinding and polishing. 2)Implement temporary visual workflow guidelines to improve coordination. 3)Schedule periodic grinding wheel maintenance to reduce uneven wear	Task	Assigned To	Deadline	Completed?	
		Define standard operating procedures (SOPs)	Process Engineer	2 Weeks	☐	
		Separate grinding & polishing roles	Production Supervisor	3 Weeks	☐	
		Implement job sequencing to minimize idle time	Production Manager	4 Weeks	☐	
		Redesign workstation layout for efficiency	Industrial Engineer	6 Weeks	☐	
		Monitor takt time vs. cycle time improvements	Quality Team	Ongoing	☐	
					☐	
3: Check	Test Results	Did your plan succeed or fail?	Areas for Improvement			
	Performance Evaluation Metrics: 1)Measure machine idle time before and after implementing job sequencing 2)Target: Reduce idle time by 20% 3)Track accumulation of parts at grinding and polishing stages. 4)Target: Reduce backlog by 30% 5) Compare actual cycle time with takt time to assess improvement 6)Target: Ensure cycle time stays within takt limits.					

Grinding & Polishing



4: Act	Will tested solution become new standard, or do	Implementation Tasks for Establishing Standard				PDCA - CYCLE PLAN DO CHECK ACT CONTINUOUS IMPROVEMENT
		Task	Assigned To	Deadline	Completed?	
		Implement separate workstations for grinding and polishing	Production Manager		☐	
		Adjust workstation layout to minimize movement waste	Industrial Engineer		☐	
		Introduce training on takt time and work balancing	Training officer		☐	
		Schedule periodic performance audits for grinding & polishing	Management Team		☐	
		Monitor machine utilization and minimize idle time	Production Supervisor		☐	
					☐	
					☐	
					☐	

Electroplating Section



The Electroplating Section at Akij Bathware is a vital stage in the production process, focused on applying metal coatings to products to improve their surface quality, durability, and aesthetics. Electroplating is primarily used to provide products with a protective metal layer, such as chrome, nickel, or gold. The section consists of three types of electroplating:

1. Auto Electroplating
2. Stainless Steel (SS) Electroplating
3. Physical Vapor Deposition (PVD) Coating

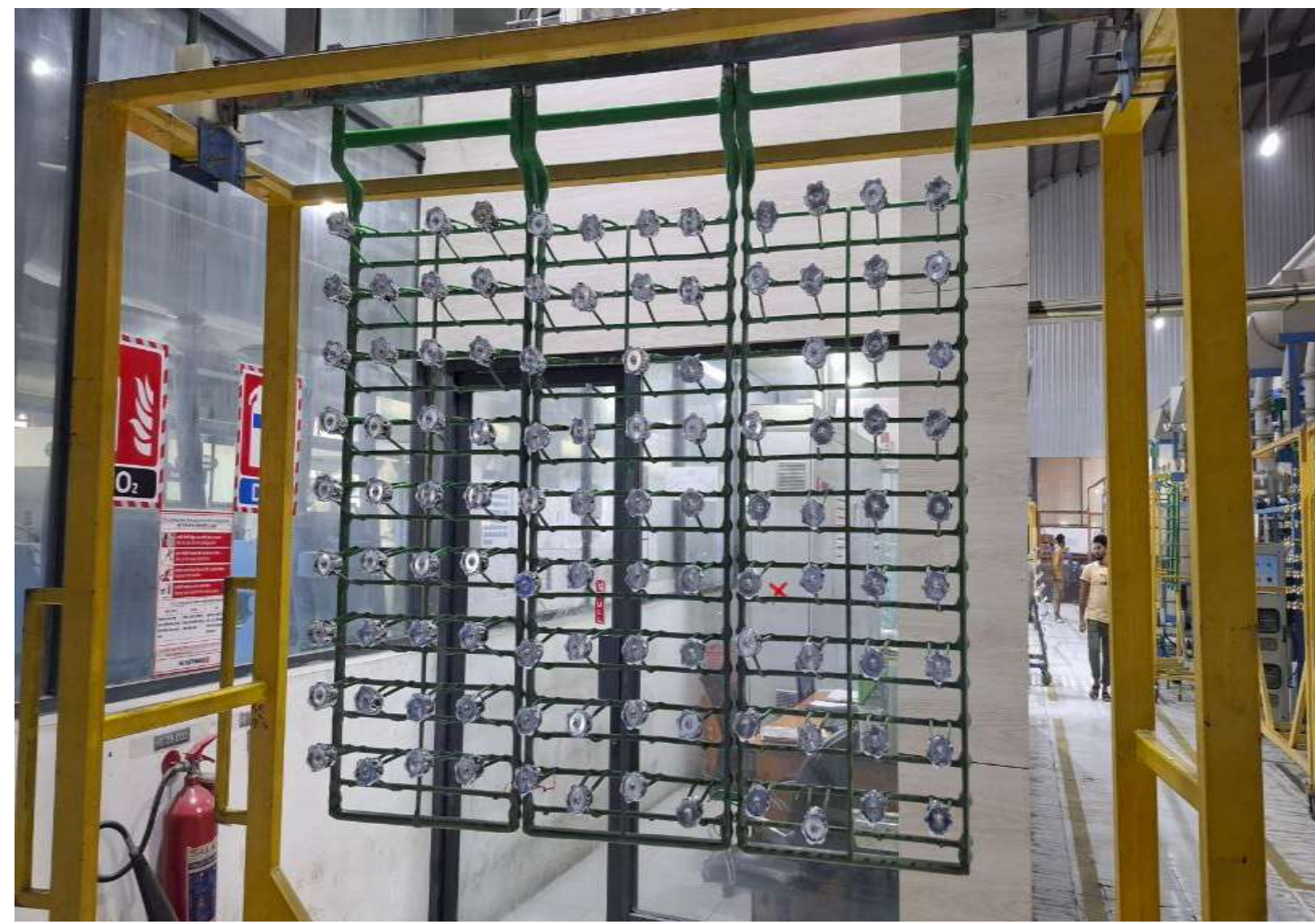


Fig: Electroplating Process in Akij Bathware Limited(Fully Automated)

Electroplating Section



**Before
Electroplating**



After Electroplating

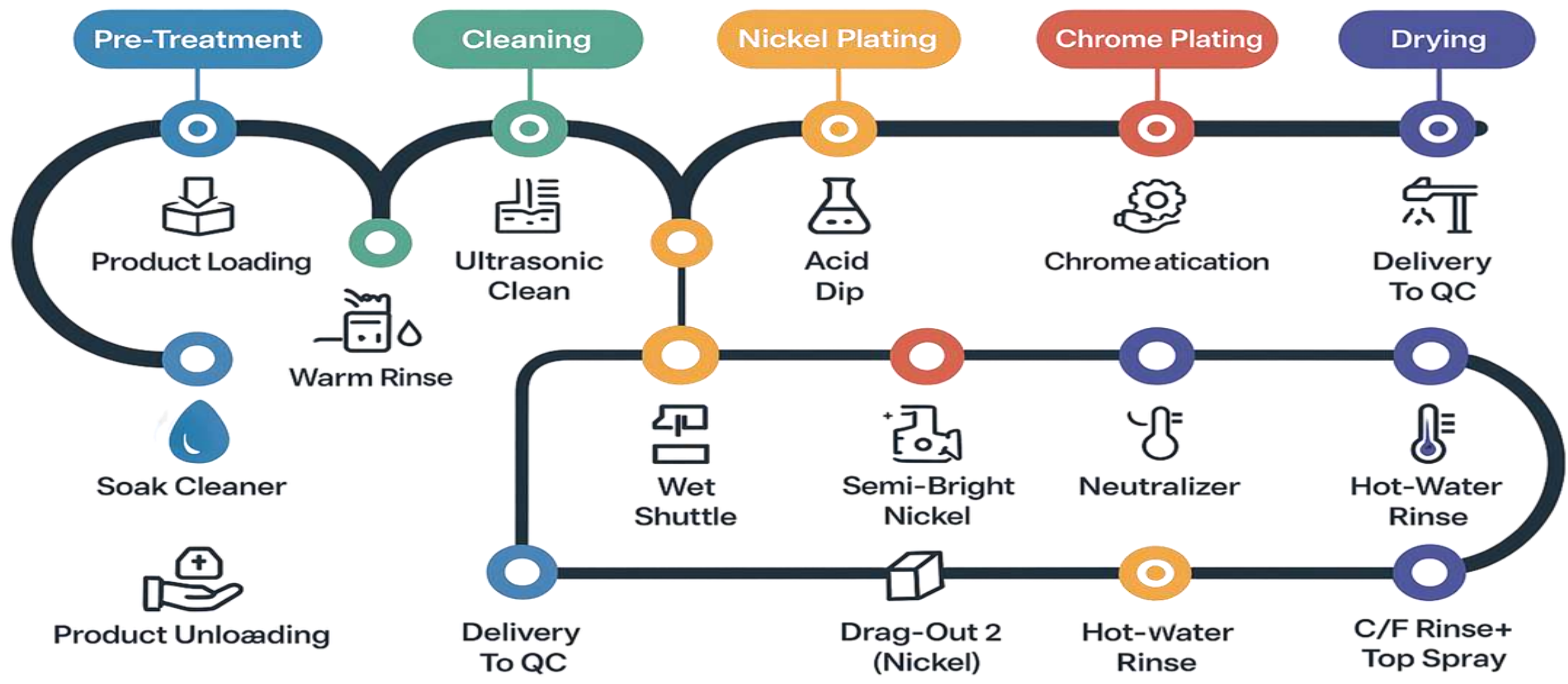
Process of Electroplating

Section



AKIJ BATHWARE

Electroplating Process



Assembly Section



The assembly section at Akij Bathware Limited is the final stage of production, where precision components from the machining, grinding, and electroplating sections are integrated into finished bathware products. This section utilizes specialized machines such as water testing machines, tightening machines, and endurance testing machines to ensure product quality, functionality, and durability before packaging and dispatch.



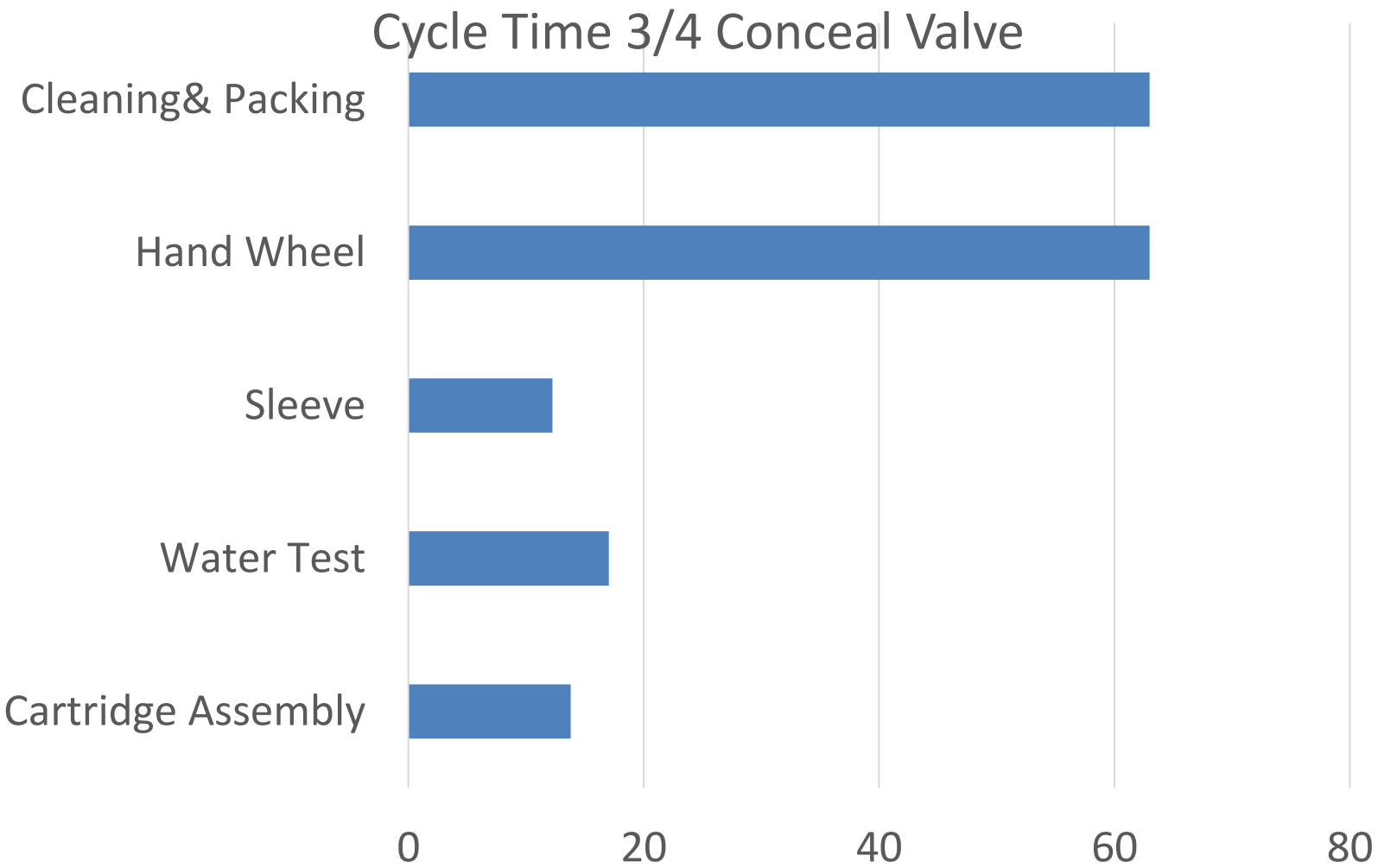
Assembly Section



Assembly Time for 3/4 Conceal valve



3/4 Conceal valve				
S.L No.	Assembly	Time Required for 5 Samples(minutes)	Cycle Time	
1	Cartridge Assembly	1.15	13.8	Seconds
2	Water Test	1.42	17.04	Seconds
3	Sleeve	1.02	12.24	Seconds
4	Hand Wheel	5.25	63	Seconds
5	Cleaning& Packing	2.1	63	Seconds
		Total	2.818	Minutes



Implementing 5S Practices Across the Entire Plant



Observation:

The plant displays signage indicating adherence to 5S principles (Sort, Set in Order, Shine, Standardize, Sustain).

No formal documentation or records were available to track the maintenance of 5S across the different sections of the plant.

Implementing 5S Practices Across the Entire Plant



Challenge Identified:

- ☐ Lack of **audit and documentation system** for ongoing 5S practices.
- ☐ No clear method to assess and sustain 5S practices plant-wide

Proposed Solution:

- ☐ **We Developed a 5S Audit Sheet** for monitoring and documentation across all plant sections (Foundry, Machining, Grinding, Electroplating, and Assembly).
- ☐ This sheet provides a structured approach to evaluate 5S implementation in each section, ensuring continuous improvement

Implementing 5S Practices Across the Entire Plant



5S AUDIT SHEET WITH RADAR CHART							
DATE: ↓	Grinding & Polishing Section	0	1	2	3	4	5
		POOR	MARGINAL	AVERAGE	GOOD	BETTER	BEST
PHASE	CRITERIA	PUT TICK (✓) MARK WHEREVER APPLICABLE					
SORT PHASE	Awareness of operators/employees about 5S Sort Phase	✓					
	Presence of Defined work instruction on 5S Sort Phase implementation						✓
	Is operators following the work instruction in implementation of 5S Sort Phase			✓			
	Free from unwanted items				✓		
	Usage of Redtag		✓				
	Proper filling of Redtag					✓	
	Maintenance of Redtag logbook						✓
	Involvement of Supervisors in sorting activity		✓				
	Involvement of Department heads in sorting activity				✓		
	Broken or worn-out tools and materials are discarded or replaced			✓			
	Proper disposal of unwanted items		✓				
	Adequate Training record on 5S Sort phase				✓		
	Obsolete safety equipment is identified and replaced						✓
	Issues in grinding and polishing are addressed through team discussions			✓			
		1	3	3	3	1	3
		0	3	6	9	4	15



Implementing 5S Practices Across the Entire Plant



SET IN ORDER PHASE	Presence of Defined work instruction on 5S Set Phase implementation				✓		✓
	Is operators following the work instruction in implementation of 5S Set Phase			✓			
	There is a proper place for each and every item				✓		
	Each place is properly labelled		✓				
	Aesthetics of work place					✓	
	Visually instructive work place						✓
		1	1	1	2	1	2
		0	1	2	6	4	10
SHINE PHASE	Awareness of operators/employees about 5S Shine Phase	✓					
	Presence of Defined work instruction on 5S Shine Phase implementation						✓
	Is operators following the work instruction in implementation of 5S Shine Phase			✓			
	Is there a cleaning schedule?				✓		
	Is cleaning done as per schedule		✓				
	Is there a record to monitor the faults identified during the inspection?					✓	
	Is there a record of action plans to eliminate the faults						✓
	PPE cleaning and maintenance are scheduled and followed			✓			
	Is the workplace have enough lighting						✓
		1	1	2	1	1	3
		0	1	4	3	4	15



Implementing 5S Practices Across the Entire Plant



STANDARDISE PHASE	Awareness of operators/employees about 5S Standardize Phase	✓					
	Presence of Defined work instruction on 5S Standardize Phase implementation						✓
	Standard operating procedures (SOPs) for grinding and polishing are documented and followed			✓			
	Uniform marking system for defect classification				✓		
	Machine downtime logs are properly maintained. Chart Area		✓				
	Regular training sessions are conducted to reinforce standardization.					✓	
		1	1	1	1	1	1
		0	1	2	3	4	5
SUSTAIN PHASE	Awareness of operators/employees about 5S Sustain Phase	✓					
	Presence of Defined work instruction on 5S Sustain Phase implementation						✓
	Is operators following the work instruction in implementation of 5S Sustain Phase			✓			
	Regular training of employees on 5S				✓		
	Regular Audit on 5S		✓				
	Involvement of Manager Chart Area					✓	
	Are employees intrested to participate in 5S	✓					
	Is sufficient time is given to Employees to participate in 5S						✓
	Presence of Knowledgeable person on 5S			✓			
	Presence of Implementation team				✓		
	Feedback from operators is collected and used for process improvements.		✓				
	Availability of pocket hand book to Employees					✓	
	Clear definition of 5S zones and competitions						✓
		2	2	2	2	2	3
		0	2	4	6	8	15
	Count each column vertically	6	8	9	9	6	12
	Multiply by the column number	0	8	18	27	24	60
	Total Score (Out of 245)	137					



Recognizing Excellence in 5S Practices



Our 5S audit sheet was introduced in the plant, and its impact has been outstanding. The plant head introduced a monthly prize-giving ceremony based on the 5S performance of each department



Implementing 5S Practices Across the Entire Plant



Increased Productivity



Higher Employee Engagement



Well-Organized Environment

Conclusion



In conclusion, the industrial attachment at Akij Bathware Limited demonstrated the transformative impact of industrial engineering tools—such as time studies, PDCA, and 5S—on manufacturing efficiency, quality, and workplace organization. System-level analysis uncovered cross-departmental dependencies, while targeted interventions resolved critical bottlenecks without major capital outlays. The adoption of preventive quality measures and lean methodologies fostered a culture of continuous improvement, underscoring the role of data-driven strategies in sustaining operational excellence. This experience affirms that well-structured industrial practice not only enhances student learning but also delivers tangible benefits to host organizations through optimized processes and empowered workforces.



»»» **THANK YOU**



Appendix

Time Study File